

Business Analytics and AI-Driven Forecasting – User Manual

Introduction

The Business Analytics and AI-Driven Forecasting System is an enterprise-grade platform designed to analyze historical business data, monitor sales activities, manage inventory, and generate predictive insights. This system uses the MERN stack (MongoDB, Express.js, React.js, Node.js) integrated with advanced Deep Learning models (LSTM, GRU) to identify sales trends, track inventory levels, and generate comprehensive predictive analytics reports.

This user manual explains how to access the web dashboard, configure the system for real-time processing, and utilize all key features for business monitoring and AI forecasting.

1. Accessing the Business Analytics Web Dashboard

Your Business Analytics application is successfully deployed on Microsoft Azure.

GitHub Repo URL:

<https://github.com/surya-12467/Business-Analytic-MERN-AI>

Dashboard Overview

The dashboard contains multiple modules to view, analyze, and manage business data:

- **Live Monitoring Module:** Real-time KPI dashboard showing total revenue, active customers, order counts, and profit percentage.
- **Sales Analytics:** View hourly, daily, and weekly sales statistics with visual charts and category-wise distributions.
- **Forecasting Reports:** Detailed breakdown of future demand and revenue predictions generated by LSTM neural networks.
- **Data Upload & Processing:** Upload CSV/Excel historical datasets for batch processing and model training.
- **Configuration Panel:** Set model parameters, forecast windows, and alert thresholds for optimal performance.

- **Alerts & Notifications:** Configure alerts for unusual transaction patterns (anomalies) or low inventory levels.
- **Export Reports:** Generate and download CSV/PDF reports of sales and forecasts for a selected time period.

2. Setting Up Data Processing

Step A: Connect Your Data Source

1. Open the Business Analytics Dashboard.
2. Navigate to **Settings > Data Sources**.
3. Choose your input method:
 - **Database Connection:** Enter MongoDB/PostgreSQL connection strings.
 - **API Feeds:** Connect external operational systems (POS, CRM, E-Commerce).
 - **File Upload:** Upload CSV, Excel, or JSON datasets.
4. Click **Connect Source**.
5. Wait for the system to initialize data ingestion (10-30 seconds for batch processing).

Your business data is now being processed for analytical extraction.

Step B: Configure Forecast Parameters

1. Click the **Settings (⚙️)** icon on the dashboard.
2. Adjust the following parameters:
 - **Forecast Window:** Set the prediction timeline (e.g., Next 7 Days, 30 Days, 3 Months).
 - **AI Model Selection:**
 - **LSTM (Recommended):** Best for complex, non-linear seasonal sales data.
 - **ARIMA:** Suitable for linear, stable time-series data.
 - **Confidence Interval:** Determine the acceptable error margin for demand predictions.
3. Click **Save Configuration**. The Python microservice will immediately apply these settings.

3. Using the Analytics System

The platform offers multiple operational modes.

A. Real-Time Dashboard Mode

1. Navigate to the **Dashboard** tab.
2. You will see:
 - Real-time counters updating total revenue, orders, and customer acquisition.
 - Live charts displaying revenue trends and profit margins.
 - Inventory status indicating current stock levels and top-selling products.
3. Click on any specific KPI card to view detailed historical comparisons.

B. Batch Data Analysis (Sales Upload)

1. Navigate to the **Upload Data** tab.
2. Click **Select File** and choose your historical dataset (CSV/Excel).
3. Configure processing options (e.g., target feature to predict, date column selection).
4. Click **Start Processing**. Monitor the progress bar as the data undergoes Min-Max Normalization and sequence windowing.
5. Once complete, the system will output clean, structured data ready for AI ingestion.

C. Analytics & Forecasting

1. Navigate to the **Forecasting** tab.
2. Select your date range and target product category using the filter parameters.
3. View automated predictive reports:
 - **Demand Trend:** Line charts comparing historical sales with AI-predicted future sales.
 - **Model Accuracy:** View metrics like MAE (Mean Absolute Error) and RMSE to gauge reliability.
4. **Export Reports:** Click the **Download Report** button to export findings in PDF or CSV formats.

4. Advanced Features

Custom Alert Configuration

1. Go to **Settings > Alerts**.
2. Create new alert rules:
 - **Rule Type: Low Inventory Alert.** Trigger when a product's stock falls below X units.

- **Rule Type: Anomaly Detection.** Automatically detect unusual spikes or drops in sales/ transactions.

3. Configure notification method (Dashboard popup, Email notification via SendGrid).

4. Click **Save Alert Rule**.

Model Re-optimization & Training

1. In the **AI Settings** panel, enable the **Auto-Retrain** toggle.
2. As new sales data flows into the MongoDB database, the Python backend will automatically retrain the LSTM models on weekends or off-peak hours to adapt to new market behaviors.

5. Troubleshooting Guide

Problem	Possible Reason	Solution
"API Connection Failed" error	Node.js backend cannot reach the Python FastAPI service.	Verify the Python microservice is running. Check Docker container logs and network bridge.
Predictions are flat or highly inaccurate	Insufficient historical data; missing dates in dataset.	Upload at least 6 months of historical data. Ensure no missing values (imputation).
Dashboard not loading / lagging	Large dataset query overloading the browser or database.	Apply date filters to reduce data load. Ensure MongoDB indexes are properly configured.
Data Upload Fails	File format not supported; file too large.	Use valid CSV or Excel formats. Ensure file size < 100MB. Check column headers.
High False-Positive Alerts	Anomaly threshold set too low.	Increase the risk threshold in the Alerts settings to reduce sensitivity.

6. System Requirements

For Web Application (End User)

- Browser: Chrome 90+, Edge 90+, Firefox 88+, Safari 14+

- Internet Connection: Minimum 2 Mbps
- Screen Resolution: 1366x768 or higher recommended
- RAM: 4GB minimum, 8GB recommended

For Deployment (Server)

- Node.js (v18+) and Python (v3.9+) environments
- Database: MongoDB cluster
- Compute: Minimum 2vCPUs, 8GB RAM (GPU recommended for faster LSTM training)

7. Technology Stack

Component	Technology
Backend	Node.js, Express.js
AI / Machine Learning	Python, FastAPI, TensorFlow/Keras, Scikit-learn
Models Used	LSTM, GRU, ARIMA, Random Forest
Frontend	React.js, TypeScript, Tailwind CSS, Chart.js
Database	MongoDB (Mongoose ODM)
Authentication	JWT (JSON Web Tokens), bcrypt
Deployment	Docker, Microsoft Azure

8. API Endpoints (For Developers)

The backend exposes REST API endpoints for integration with external systems:

Upload Sales Data Endpoint

```
POST /api/upload_sales
Request Body: multipart/form-data (CSV/Excel file)
Response:
{
  "status": "success",
  "records_processed": 1450,
```

```
"message": "Data successfully ingested into MongoDB"
}
```

Predictive Inference Endpoint (via Python Microservice)

```
POST /api/predict
Request Body:
{
  "product_category": "Electronics",
  "forecast_days": 7,
  "model": "LSTM"
}
Response:
{
  "predictions": [
    {"date": "2026-06-18", "predicted_sales": 125, "revenue": 45000},
    {"date": "2026-06-19", "predicted_sales": 140, "revenue": 52000}
  ],
  "rmse": 112.35
}
```

9. Best Practices & Tips

1. **Data Hygiene:** Ensure uploaded CSV files have consistent column headers and date formats (YYYY-MM-DD) for accurate ingestion.
2. **Model Selection:** Use "LSTM" for datasets with strong seasonal trends or complex patterns. Use "ARIMA" only for simple, predictable data.
3. **Performance Optimization:** If the dashboard lags, avoid querying the entire database history. Use the date-picker to analyze data in 30-day or 90-day chunks.
4. **Regular Retraining:** Business dynamics change. Enable auto-retrain or manually retrain the AI models every month using the latest sales data.
5. **Security:** Never share your JWT tokens or API keys. Ensure role-based access is correctly assigned to new employees.

10. Contact & Support

For technical support, access issues, or feature requests:

- **Email:** suryapdaur199@gmail.com / kajimohsin65@gmail.com
- **GitHub Issues:** <https://github.com/surya-12467/Business-Analytic-MERN-AI/issues>

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Version: 1.0

Compatibility: Business Analytics Platform v1.0+